

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459226.9206 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.016
Transit depth (Rp/Rs)^2	2.56 +/- 0.5 [%]
Orbital Inclination (inc)	81.22 +/- 1.34
Airmass coefficient 1 (a1)	1.063 +/- 0.01
Airmass coefficient 2 (a2)	-0.04 +/- 0.031
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	#1 - [301, 59]
Optimal Method	PSF photometry
Transit Duration (day)	0.0473 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.16 ± 0.016 vs 0.1594 (deviation 0.03σ)
Duration fit vs published	0.0473 d vs 0.0512 d (deviation 0.68σ)
Mid-time O–C	-1.7 min
Depth SNR	9.2
Residual scatter	0.98 %

Light curve

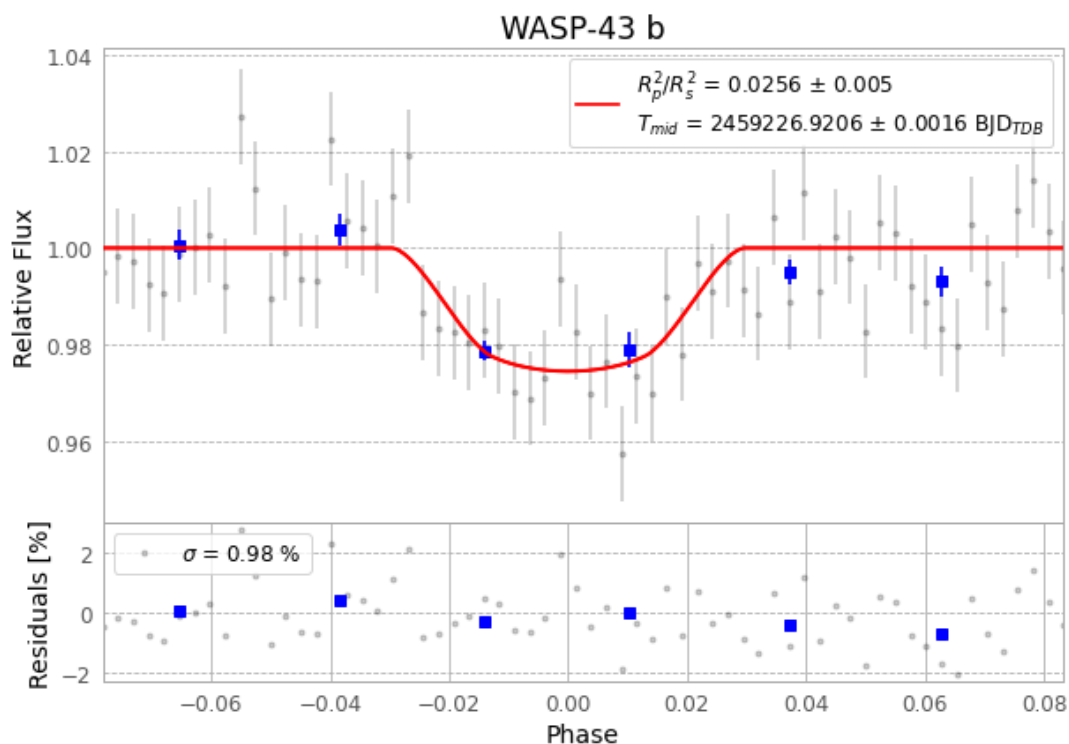


Figure 1 — FinalLightCurve_WASP-43 b_2021-01-12.png (SHA-256 7e141d3b18344fbc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ab41d2aeae769872...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43210112082712.FITS → WASP-43210112113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210112.log (6067f42f8ebe...), FinalLightCurve_WASP-43 b_2021-01-12.png (7e141d3b1834...), FinalLightCurve_WASP-43 b_2021-01-12.csv (3316d5e1a521...)

Compute resources

Wall time 162.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:55Z.